



LINAC Access Control Interlock System Validation and Verification Report – Oct 2007

1.5.52.4 Rev. 0

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REVISION HISTORY

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1.0 PURPOSE

This report outlines the results from Validation and Verification (V&V) of the LINAC Access Control Interlock System (ACIS) conducted from October 22 – October 24, 2007.

2.0 BACKGROUND

The validation and verification of the LINAC ACIS ensures that it is operating within the design specifications outlined in “1.2.37.2.Rev.0-Personnel Safety Lockup System-West [1] and in the 1.2.37.3.Rev.0-Personnel Safety Lockup System Micro84 Program-West [2]. The V&V procedure also verifies the “LINAC-LTB1 Lockup Procedure” 2.7.37.2 Rev.2 [3].

3.0 ACRONYMS

ACIS	Access Control Interlock System
BST	Beam Stop
EOS	Emergency Off Station
EXB	Exit Button
HRN	Horn
HSE	Health Safety and Environment Department
HV	High Voltage
H/W	Hard Wired
LOTO	Lock Out Tag Out
MKS	MKS Integrity Client Problem Tracking System
RCC	Radiological Control Coordinator
V&V	Validation and Verification
ZLL	Zone Lockup Lights

4.0 VERIFICATION AND VALIDATION RESULTS

4.1 SUMMARY

Two non-conformances were identified during the V&V testing and were entered into the MKS problem tracking system.

1) MKS 2370

During the annual V&V testing of the LINAC, the following lights were found to be not working when tested.

Zone 1 by LUS 4
Zone 2 across from LUS 5
Zone 4 nearest to LUS 1
Zone 5 in rm. 0108 (area 5.1)

The lights were replaced, tested and found to be working properly.

2) MKS 2408

During ACIS V&V in zone 4 of the LINAC, the inspectors observed that the light bulb at LUS1 in the Kirk Key box was not working. The area could however be successfully locked up.

The light bulb in the Kirk Key Lock box was changed, tested, and found to be working properly.

All checklists and the non-conformance forms are attached.

Micro84 checks were undertaken separately by HSE staff in February and again in August of 2007. No problems were encountered. Checklists for the Micro84 ACIS V&V are kept in a binder in room 1021.

5.0 REFERENCE MATERIALS

[1] Personnel Safety Lockup System-West 1.2.37.2.Rev.0

[2] Personnel Safety Lockup System Micro84 Program-West 1.2.37.3.Rev.0

[3] LINAC-LTB1 Lockup Procedure" 2.7.37.2 Rev.2

6.0 ATTACHMENTS

LINAC Validation and Verification Checklists

Non-conformance reports (MKS 2370, 2408)

ATTACHMENTS FOR

**1.5.52.4 Rev. 0 – LINAC Access Control Interlock System Validation
and Verification Report – Oct 2007**

Available with original, signed-off document